Evaluating the health of City of Toronto's street trees by taxonomy and size

INF2167 - R for Data Science

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Abstract

Introduction

Relevant Literature

Dataset & Methods

Dataset Source

The street tree dataset used for this project is sourced from Toronto Open Data (City of Toronto 2026), a publicly available data initiative maintained by the City of Toronto. This dataset was published by City of Toronto's Parks, Forestry, and Recreation division initially on December 1, 2011 and was last updated on June 4, 2026.

The original dataset contains 16 variables, focusing on unique IDs, geographic location, tree type, and tree size. There is a total of 688,335 rows in the dataset.

Ward names were sourced from Toronto Open Data's Ward Profiles dataset (City of Toronto, City Planning 2024)

Data Cleaning

A new dataset variable was created, called tree_data. This dataset only contains variables that were necessary for this project, including:

- OBJECTID, a unique ID of each tree;

- WARD, the tree's location by geographical district;
- BOTANICAL_NAME, the tree's formal scientific name;and
- DBH_TRUNK, the tree's diameter at breast height.

Another new dataset was created, called tree_data_clean. The janitor package (Firke 2023) was used to clean up the variables names in the dataset so that names were in a consistent snake case format. A new "region" variable was created based on the existing "ward" variable by categorizing wards by region. For example, the trees in Wards 6, 8, 15, 16, 17, and 18 were categorized as being located in North York.

One of the focuses for this project is the 10-20-30 rule which requires the classification of the trees by species, genus, and family. Though species and genus may be inferred for some trees based on botanical name (e.g., if the botanical name is Acer Platanoides, the species is Acer Platanoides and the genus is Acer), the tree's family name must be identified using an external source. Thus, the World Flora Online (WFO) package [WorldFlora2020] was used to create the variables, species_name, genus_name, and family_name, based on the existing variable, botanical name. To optimize runtime, a new variable, called unique_trees, was created to identify all unique botanical names in the tree_data_clean dataset. This table had a total of 304 unique tree names. By running the WFO package's classification function on the unique_trees table instead of the tree_data_clean dataset, the function had to match 304 unique botanical names instead of matching names for each of the 688,335 tree entries individually, saving processing power and optimizing time efficiency.

After the botanical names in unique_trees were classified (as the variable, taxonomy_extract), new variables in the tree_data_clean dataset were created for species_name, genus_name, and family_name. The left_join() function (Wickham et al. 2019) was used to join the species, genus, and family names by the botanical_name variable present in both datasets. The results of the WFO matching were checked for missing values, and there were a total of 18 botanical names that the classification step was unable to classify (see Table 1).

Table 1: Botanical names unidentified by WFO classification

botanical_name	n
Acer x freemanii (A. rubrum x saccharinum) 'Autumn Blaze'	15025
Quercus alba	3807
Acer x freemanii (A. rubrum x saccharinum) 'Armstrong'	2781
Magnolia x soulngeana (M. denudata x liliiflora)	2688
Acer x freemanii (A. rubrum x saccharinum) 'Marmo'	1000
Acer x freemanii (A. rubrum x saccharinum) 'Celebration'	965
Acer x freemanii (A. rubrum x saccharinum) 'Sienna Glen'	837
Acer x freemanii (A. rubrum x saccharinum)	563
Salix x sepulcralis 'Chrysocoma'	545
Ulmus x hollandica (U. glabra x minor) 'Pioneer'	425
Populus x canadensis (Populus nigra x deltoides)	249
Tilia x euchlora (T. cordata x dasystyla)	151
Prunus x cistena (P. pumila x P. cerasifera 'Atropurpurea')	133
alix × sepulcralis	101

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botanical_name	n
None	56
Prunus x yedoensis 'unknown hybrid'	31
Amelanchier x grandiflora (A. arborea x laevis) 'Robin Hill'	15
Sorbus aria	4

The missing data was resolved manually by identifying the species, genus, and family names for botanical names that appear more than 1000 times in the tree_data_clean dataset. Missing classifications for botanical names with less than 1000 trees were considered insignificant. Manual identification was used for Acer x freemanii, Quercus alba, and Magnolia x soulngeana. Acer x freemanii made up a significant portion of Toronto's street tree population with a total of 21,171 trees.

Methods

Exploratory data analysis was conducted to evaluate ward-level compliance to the 10-20-30 rule. Linear regression modelling was then used to investigate the relationship between the 10-20-30 rule compliance and trunk size (DBH) distributions in wards.

Results & Findings

Exploratory Data Analysis

Exploratory data analysis was conducted to understand the general data patterns. The street tree data was grouped by ward to identify the dominant species, genus, and family of each ward (see Table 2). It was found that most wards had the same pattern in which the dominant species, genus, and family were Acer platanoides, Acer, and Sapindaceae respectively. The proportions of the largest species, genus, and family were then calculated. It was found that all wards violated the <10% species rule; 23 out of 25 wards violated the <20% genus rule; and 8 out of 25 wards violated the <30% family rule.

Table 2: Dominant species, genus, and family names and proportions per ward

ward	ward_name	max_species	dominant_species	max_genus	dominant_genus	max_family	dominant_family
1	Etobicoke North	0.1635295	Acer platanoides	0.2383636	Acer	0.2589396	Sapindaceae
2	Etobicoke Centre	0.1825712	Acer platanoides	0.2985332	Acer	0.3193174	Sapindaceae

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ward	ward_name	max_species	dominant_species	max_genus	dominant_genus	max_family	dominant_family
3	Etobicoke-Lakeshore	0.1328476	Acer platanoides	0.2896189	Acer	0.3067255	Sapindaceae
4	Parkdale-High Park	0.1199773	Acer platanoides	0.2569476	Acer	0.2715687	Sapindaceae
5	York South-Weston	0.1276188	Acer platanoides	0.2645255	Acer	0.2817541	Sapindaceae
6	York Centre	0.1131337	Acer platanoides	0.2096812	Acer	0.2277967	Sapindaceae
7	Humber River-Black Creek	0.1054473	Acer platanoides	0.1820105	Acer	0.2014564	Sapindaceae
8	Eglinton-Lawrence	0.1418734	Acer platanoides	0.2719099	Acer	0.2861086	Sapindaceae
9	Davenport	0.1148122	Acer platanoides	0.2606241	Acer	0.2724800	Sapindaceae
10	Spadina-Fort York	0.1252920	Gleditsia triacanthos	0.2145020	Acer	0.2347495	Sapindaceae
11	University-Rosedale	0.1323976	Acer platanoides	0.2868825	Acer	0.3042910	Sapindaceae
12	Toronto-St. Paul's	0.1634760	Acer platanoides	0.2879129	Acer	0.2976351	Sapindaceae
13	Toronto Centre	0.1648532	Gleditsia triacanthos	0.1945712	Gleditsia	0.2411372	Fabaceae
14	Toronto-Danforth	0.1456086	Acer platanoides	0.3193180	Acer	0.3362101	Sapindaceae
15	Don Valley West	0.1701510	Acer platanoides	0.2906350	Acer	0.2998805	Sapindaceae
16	Don Valley East	0.1530587	Acer platanoides	0.2586868	Acer	0.2688010	Sapindaceae
17	Don Valley North	0.1736035	Acer platanoides	0.2707713	Acer	0.2842581	Sapindaceae
18	Willowdale	0.1373216	Acer platanoides	0.2731174	Acer	0.2885025	Sapindaceae
19	Beaches-East York	0.1264929	Acer platanoides	0.2830332	Acer	0.2965877	Sapindaceae
20	Scarborough Southwest	0.1154172	Acer platanoides	0.2658264	Acer	0.2853447	Sapindaceae
21	Scarborough Centre	0.1250248	Acer platanoides	0.2531231	Acer	0.2755800	Sapindaceae
22	Scarborough-Agincourt	0.2018939	Acer platanoides	0.3152064	Acer	0.3413165	Sapindaceae

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ward	ward_name	max_species	dominant_species	max_genus	dominant_genus	max_family	dominant_family
23	Scarborough North	0.1816069	Acer platanoides	0.2976157	Acer	0.3317171	Sapindaceae
24	Scarborough-Guildwood	0.1434368	Acer platanoides	0.2731563	Acer	0.3005028	Sapindaceae
25	Scarborough-Rouge Park	0.1634354	Acer platanoides	0.3169567	Acer	0.3351326	Sapindaceae
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In addition to ward-level compliance to the 10-20-30 rule, Table 3 was created as a summary of the size distribution of each ward, including number fo trees, mean diameter (DBH), median diameter (DBH), and standard deviation (DBH).

Table 3: Summary of size distributions per ward

ward	ward_name	total_trees_ward	mean_dbh	median_dbh	sd_dbh
1	Etobicoke North	37082	20.31465	11	19.55837
2	Etobicoke Centre	52155	27.62345	23	22.83915
3	Etobicoke-Lakeshore	46941	27.20762	19	30.76771
4	Parkdale-High Park	28247	28.94775	20	27.34940
5	York South-Weston	28209	24.37513	18	22.09037
6	York Centre	34004	21.17298	14	18.80286
7	Humber River-Black Creek	24581	17.79952	10	16.61476
8	Eglinton-Lawrence	35144	26.94986	20	25.35631
9	Davenport	19484	23.25549	15	22.25355
10	Spadina-Fort York	11557	21.57177	14	19.58469
11	University-Rosedale	23724	25.31011	16	24.05542
12	Toronto-St. Paul's	26640	26.39790	18	25.25155
13	Toronto Centre	8547	22.46098	15	19.69410
14	Toronto-Danforth	19003	27.42788	20	26.31245
15	Don Valley West	40993	30.17276	25	51.63810
16	Don Valley East	24520	25.77276	20	19.84693
17	Don Valley North	31883	25.91187	22	20.14952
18	Willowdale	31459	27.69465	23	22.49197
19	Beaches-East York	21100	27.42261	20	26.03292
20	Scarborough Southwest	21262	24.16758	13	24.46260
21	Scarborough Centre	20172	21.00570	8	21.51679

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ward	ward_name	total_trees_ward	mean_dbh	median_dbh	sd_dbh
22	Scarborough-Agincourt	21754	20.83654	12	19.76509
23	Scarborough North	24955	18.34847	10	15.04367
24	Scarborough-Guildwood	22672	21.61547	10	21.46700
25	Scarborough-Rouge Park	27234	20.74007	14	18.59513
*					

Linear Regression Modelling

Two linear regression models were created to compare relationships between the 10-20-30 rule and tree size distributions. Model 1 was average diameter as a model of the percentage proportion of a ward's dominant species, genus, and family. Alternatively, Model 2 was average diameter as a model of the excess percentage points of a ward's dominant species, genus, and family. Comparing the models, Model 2 had higher R-squared and adjusted R-squared values, indicating a stronger model fit, and lower p-value and residual standard error, indicating higher statistical significance and lower error (see Table 4).

Table 4: Comparison of Models 1 and 2

Metric	Model 1	Model 2
R-Squared	0.33120	0.56590
Adjusted R-Squared	0.24000	0.50670
p-value	0.02874	0.00031
Residual Std. Error	3.08600	2.48600

Taking a look at Model 2's results in Table 5, the species term has a p-value greater than 0.05, indicating that it is statistically insignificant. However, both genus and family are statistically significant. The model demonstrates that:

- for every 1 percentage point a ward exceeds the 20% genus threshold, the average trunk diameter increases by 0.90 cm; and
- for every 1 percentage point a ward exceeds the 30% family threshold, the average trunk diameter decreases by ~2.03 cm.

The results of Model 2 are shown in Figure 1.

Table 5: Model 2 Results

term	estimate	std.error	statistic	p.value
(Intercept)	19.2639812	1.3516386	14.2523167	0.0000000
species_excess	0.0519455	0.2037781	0.2549119	0.8011590
genus_excess	0.9001926	0.1970826	4.5675909	0.0001508
family_excess	-2.0302680	0.5028253	-4.0377207	0.0005501

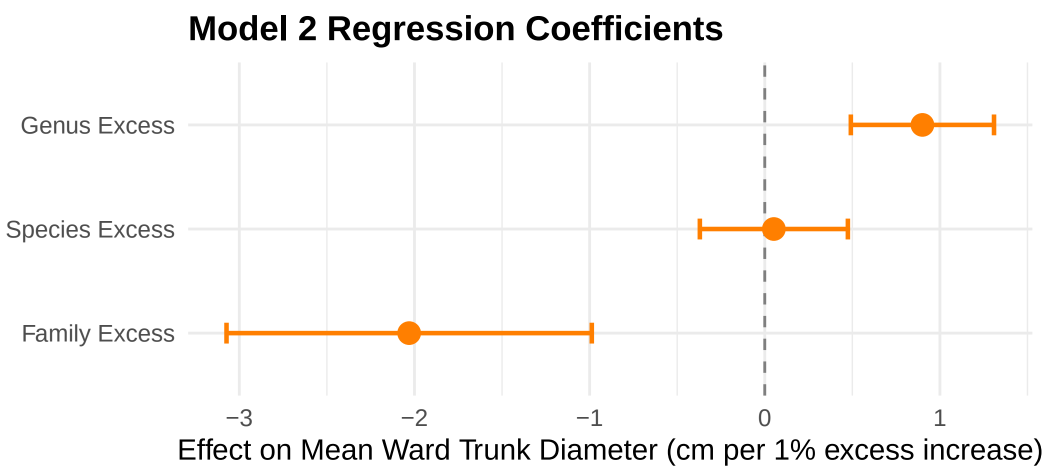

Figure 1: Model 2 regression coefficients by species, genus, and family

Discussion

Based on the relationship between average diameter and excess genus percentage points, Model 2 shows that wards with severe violation of the 20% genus rule rely heavily on larger, mature trees in their street tree population. This reflects the historical urban forestry practice of over-planting Acer, more commonly known as Maple trees.

The trend of over-planting Acer trees is also evident in Figure 2. There are Acer trees of varying sizes, indicating that there are many older Acer trees that have survived to maturity as well as many young Acer trees that have been recently planted. In comparison, other genera have few to no mature trees, which could be explained by lower survival rates due to harsh weather and road conditions.

Since other genera have low survivial rates, it is possible that the city has opted to plant more Acer trees knowing that Acer trees are more resilient to severe environmental conditions and have higher likelihood to reach maturity.

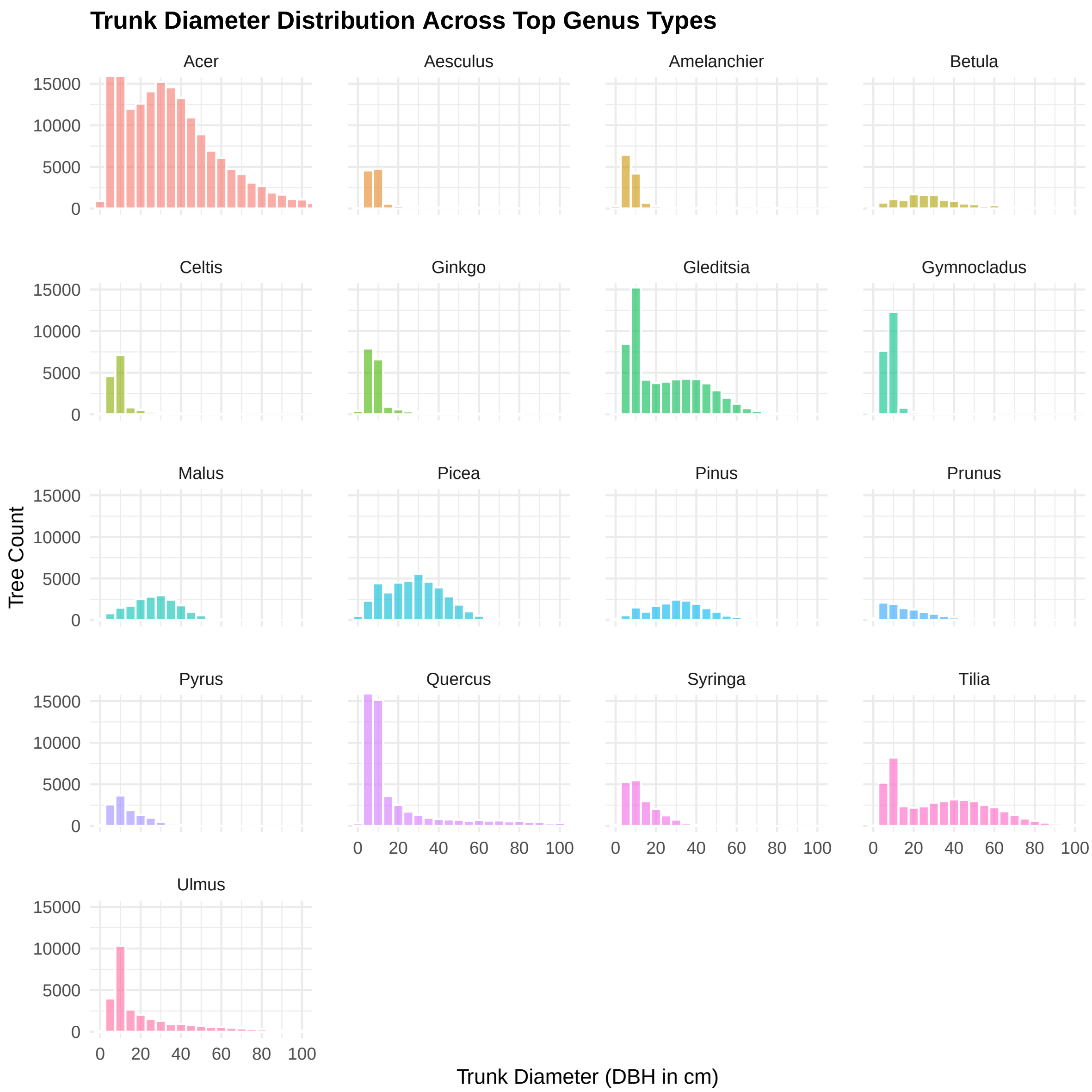

Figure 2: Size distribution of most planted genera

Conversely, Model 2 found that exceeding the 30% family threshold is associated with significantly smaller average trunk diameters. Though the Acer genus belongs within the Sapindaceae family, this pattern can be explained as recent mass planting of Sapindaceae trees that do not belong under the Acer genus. In Figure 3, there are multiple families with recent plantings, such as fabaceae, fagaceae, and rosaceae. However, sapindaceae still holds the largest proportion of the young tree population.

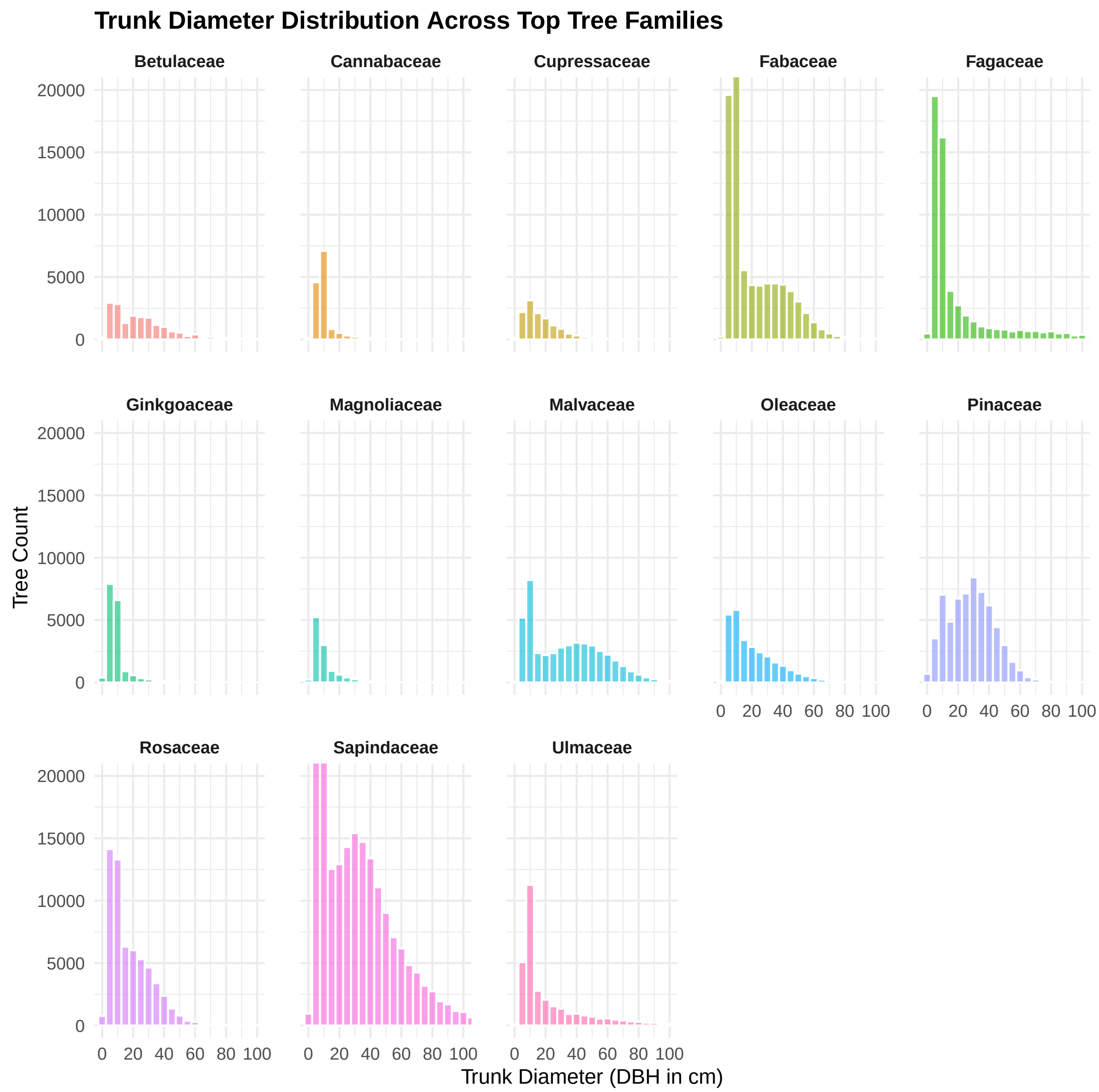

Figure 3: Size distribution of most planted families

Conclusion

This study evaluated the health of Toronto's street tree population by assessing adherence to the 10-20-30 rule against trunk diameter distributions across 25 municipal wards. The analysis confirmed severe taxonomic over-concentration, driven by a legacy of over-planting Acer species.

Achieving a sustainable and resilient urban canopy will require the City of Toronto not only to restrict further over-planting of dominant genera, but also to prioritize long-term care and survival strategies for non-dominant species as they grow toward maturity.

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